

Validating a Gamified Selective Attention Task: A Non-Parametric Analysis of Concurrent Validity, Target Load, Modality, and Game Level Effects

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1 Introduction

Selective attention - the capacity to focus on task-relevant stimuli while suppressing irrelevant distractors - is central to nearly every goal-directed activity humans perform (Treisman & Gelade, 1980). Its efficiency depends critically on *target load*: when multiple targets must be tracked simultaneously, focused attention must be serially deployed, creating a cognitive bottleneck that increases reaction times with load (Treisman & Gelade, 1980). There is growing interest in gamified cognitive assessment as a means of measuring attentional capacity in ecologically valid settings (Lumsden et al., 2016), but any novel instrument must first be *validated* against an established gold-standard paradigm.

1.1 Dataset and Experimental Design

This study used a 2×2 Mixed Factorial Design to validate a newly developed Selective Attention Game against a standard Lab Task. A total of 37 participants took part in the experiment. The design included one between-subjects factor and one within-subjects factor. The between-subjects factor was Target Load, with two levels: Single Target and Multiple Target. Participants were randomly assigned to only one of these target-load groups. Group A consisted of 21 participants who completed the Single Target condition. Group B consisted of 16 participants who completed the Multiple Target condition. Participants in Group A performed both the Single Target Game and the Single Target Lab Task. Participants in Group B performed both the Multiple Target Game and the Multiple Target Lab Task. The within-subjects factor was Modality, which included Game and Lab conditions. Because modality was within-subjects, every participant completed both the game and lab versions. This allowed comparison of performance within the same participant across modalities. The study also used counterbalancing to reduce practice and fatigue effects. Half of the participants completed the game first, while the other half completed the lab task first. This design enabled researchers to examine the effects of target load, modality, and their interaction on selective attention performance.

The study employed a 2×2 Mixed Factorial Design:

- **Factor 1 - Target Load (Between-Subjects)**
- **Factor 2 - Modality (Within-Subjects)**

The design yields four experimental cells: *Single Lab*, *Single Phone or Game*, *Multi Lab*,

and *Multi Phone or Game*. The primary dependent variable is **Reaction Time (RT)** in milliseconds, with Accuracy (percentage correct) as a secondary measure. The key research questions are:

1. **RQ1 — Concurrent Validity:** Does game RT correlate positively with lab RT, confirming that both measure the same underlying construct?
2. **RQ2 — Target Load Effect:** Is performance significantly worse in the Multiple Target condition relative to the Single Target condition?
3. **RQ3 — Modality Effect:** Does the gamified smartphone interface alter performance relative to the standard laboratory interface?
4. **RQ4 — Game Level Effect:** Does RT change systematically across increasing game difficulty levels, indicating a learning or fatigue trajectory?

2 Methods

2.1 Data Preparation and Preprocessing

Raw trial-level data were pre-aggregated into participant-level mean RT and accuracy scores, yielding two CSV files (`participant_rt_by_group.csv`, `participant_accuracy_by_group.csv`; 37 rows $\times$ 4 columns each). Wide-format files were reshaped via `pd.melt()` and merged into a unified long-format dataframe of 74 rows (37 $\times$ 2 modalities). For RQ4 and regression analyses, full trial-level game data (480 observations, up to 15 levels) were loaded separately.

2.2 Outlier Detection

Outliers were flagged using the IQR method ($< Q_1 - 1.5 \text{ IQR}$ or $> Q_3 + 1.5 \text{ IQR}$) within each condition, replacing values with `NaN` (4 RT; 5 Accuracy outliers removed). All hypothesis tests are reported pre- and post-removal to assess robustness.

2.3 Normality Testing and Test Selection

Shapiro-Wilk tests were applied to each condition ($\alpha = .05$). Four of eight condition-DV combinations violated normality: Single Lab RT ($W = 0.869$, $p = .009$), Multi Lab RT ($W = 0.695$, $p < .001$), Single Game Accuracy ($W = 0.762$, $p < .001$), Multi Lab Accuracy ($W = 0.406$, $p < .001$). The overall RT distribution (skewness = 1.12; Figure 3a) further confirmed right-skew typical of RT data. Non-parametric tests were therefore applied to RQ1–RQ4 for consistency. Log-transformation normalised RT for Mixed ANOVA and OLS regression (all log-RT Shapiro-Wilk $p > .36$). The exception is Analyses 8–9 (paired t -tests), where difference-score normality was confirmed (Multiple Phone: $W = 0.930$, $p = .335$; Single Phone: $W = 0.972$, $p = .798$).

Test selection summary: Spearman’s ρ (RQ1); Mann-Whitney U (RQ2); Wilcoxon Signed-Rank (RQ3); Friedman test (RQ4); 2 $\times$ 2 Mixed ANOVA on log(RT); OLS on log(RT) with Load, Modality, interaction; OLS on log(RT) with Level $\times$ Load; paired t -tests (Analyses 8–9). All analyses conducted in Python (v3.11) using `scipy`, `statsmodels`, `pingouin`, `pandas`; $\alpha = .05$.

3 Results

3.1 Exploratory Data Analysis

Table 1 reports descriptive statistics for all four conditions. Single Game RT ($M = 2958$ ms, $SD = 521$ ms) is dramatically higher than all other conditions. The Multiple target group is faster in both modalities (Multi Lab: $M = 1356$ ms; Multi Game: $M = 1191$ ms), which is counter-intuitive and explored in RQ2. Accuracy is near-ceiling across conditions; Single Lab achieves a perfect ceiling ($M = 100\%$, $SD = 0$).

Table 1: Descriptive Statistics - Reaction Time and Accuracy (Participant-Level Means)

Condition	DV	N	M	SD	SE	Median	Range
Single Lab	RT (ms)	21	1565.41	236.54	51.62	1540.26	1160–2331
Single Lab	Accuracy (%)	21	100.00	0.00	0.00	100.00	100–100
Single Game	RT (ms)	21	2957.55	520.82	113.65	2887.47	2117–3968
Single Game	Accuracy (%)	21	95.79	5.56	1.21	96.88	82–100
Multi Lab	RT (ms)	16	1356.35	216.98	54.25	1331.07	1160–2087
Multi Lab	Accuracy (%)	16	94.83	13.39	3.35	98.67	45–100
Multi Game	RT (ms)	16	1191.45	236.89	59.22	1202.65	811–1815
Multi Game	Accuracy (%)	16	96.37	2.53	0.63	96.35	91–100

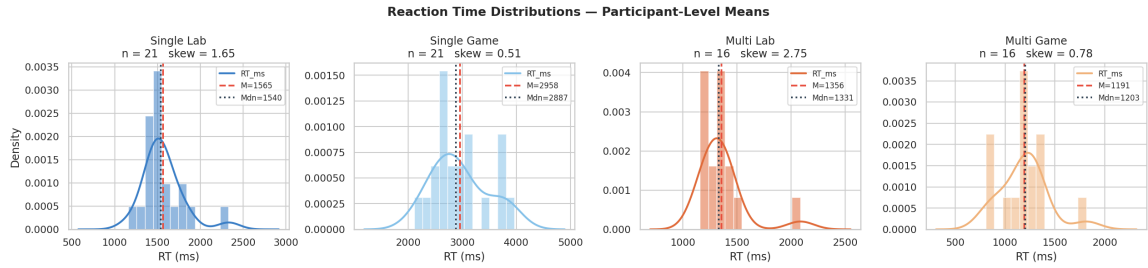


Figure 1: RT distributions (participant-level means) for all four conditions. Dashed red = mean; dotted black = median. Skewness: Single Lab 1.65, Multi Lab 2.75, Single Game 0.51, Multi Game 0.78.

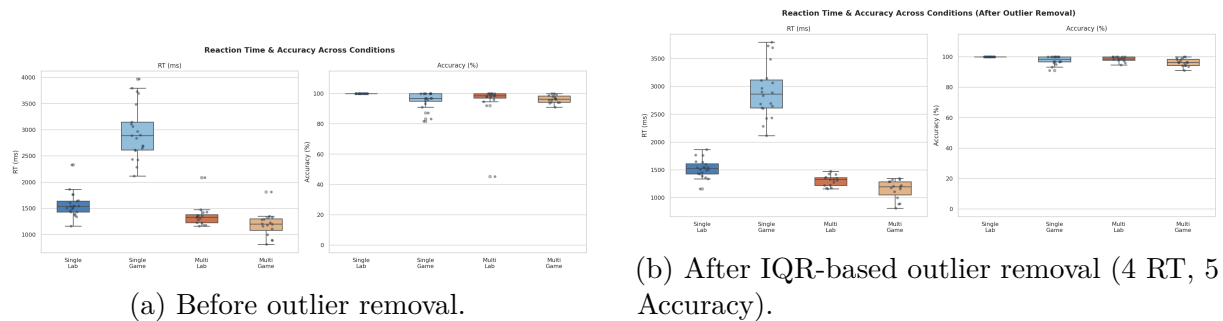


Figure 2: Boxplots of RT and Accuracy across conditions.

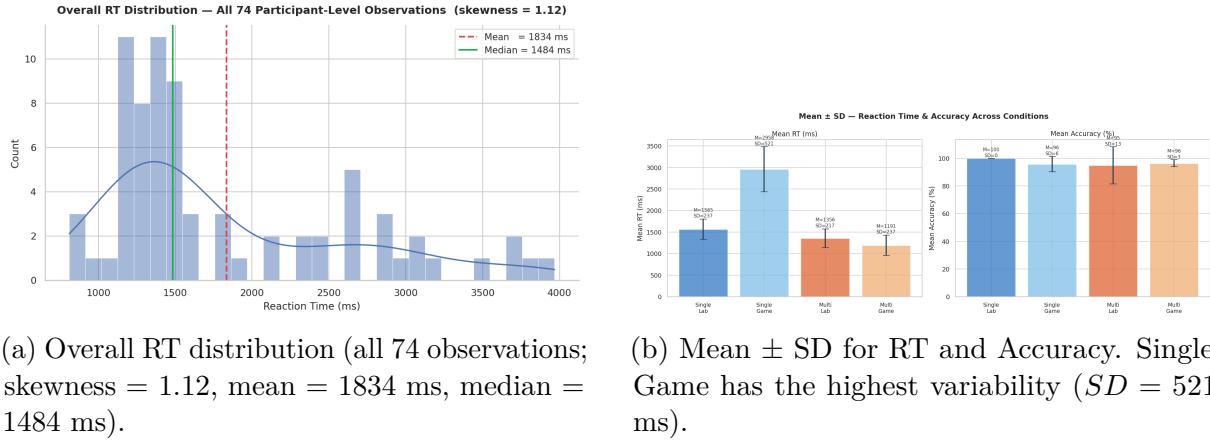
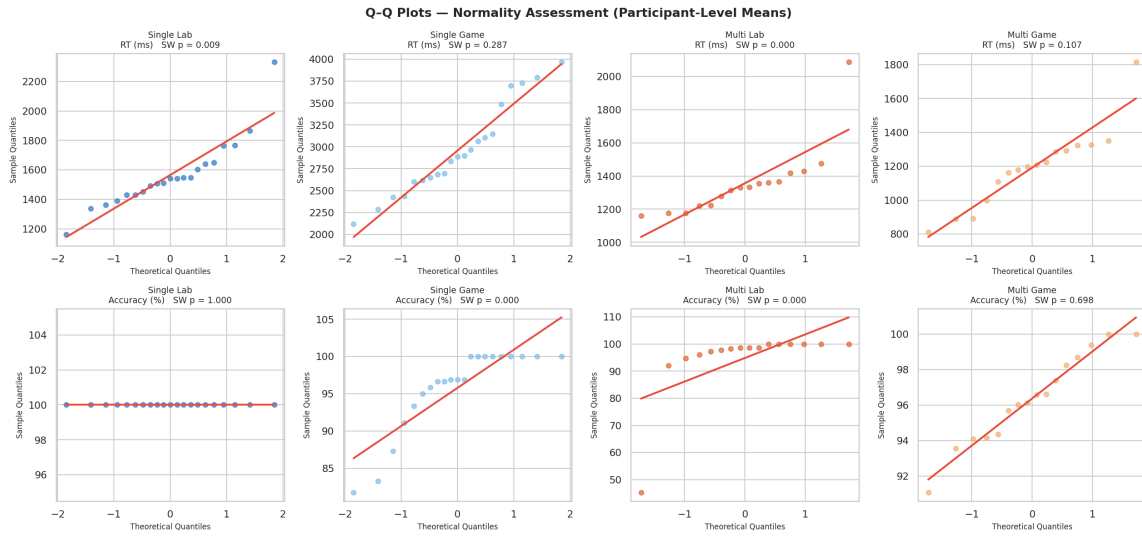


Figure 3: Overall RT distribution and condition-level summary statistics.

Figure 4: Q-Q plots (Shapiro-Wilk p -values in subtitles). Single Lab RT ($p = .009$), Multi Lab RT ($p < .001$), Single Game Accuracy ($p < .001$), Multi Lab Accuracy ($p < .001$) violate normality.

3.2 Analysis 1 - RQ1: Concurrent Validity (Spearman's ρ)

Spearman's ρ was used (non-normal distributions; tests monotonic association). Results are reported per target-load group.

Single Target ($n = 21$): $\rho = 0.396$, $p = .076$, 95% CI $[-0.043, 0.707]$. Moderate positive correlation but non-significant. **Fail to Reject H_0 .**

Multiple Target ($n = 16$): $\rho = 0.035$, $p = .897$, 95% CI $[-0.469, 0.522]$. Negligible, non-significant. **Fail to Reject H_0 .**

Combined ($n = 37$): $\rho = 0.621$, $p < .001$, 95% CI $[0.372, 0.787]$. Large and significant, but likely a Simpson's Paradox artefact inflated by the between-group RT scale difference. **Reject H_0 .**

After outlier removal: Single ($n = 19$) $\rho = 0.339$, $p = .156$; Multiple ($n = 14$) $\rho = 0.015$, $p = .958$; Combined ($n = 33$) $\rho = 0.707$, $p < .001$. Pattern is consistent - within-group validity unsupported.

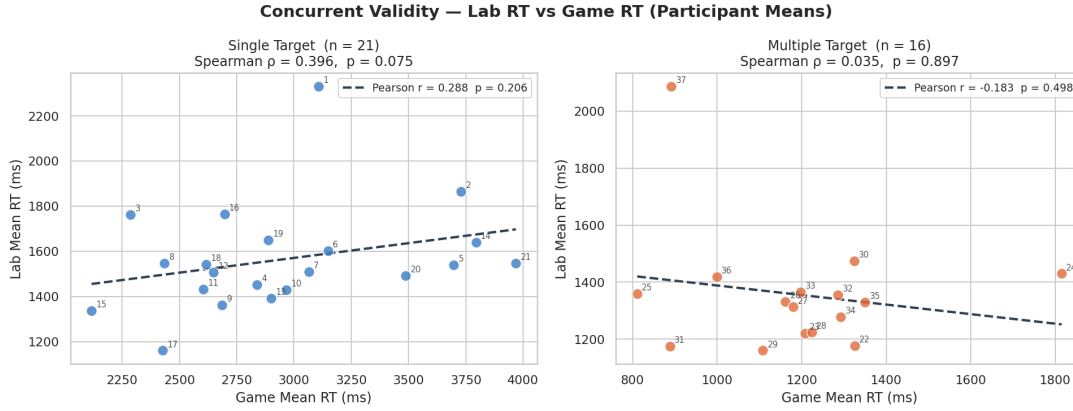


Figure 5: Concurrent validity: Lab Mean RT vs. Game Mean RT for Single (left) and Multiple (right) target groups. Spearman ρ and Pearson r shown in subtitles.

3.3 Analysis 2 - RQ2: Target Load Effect (Mann-Whitney U)

Mann-Whitney U was used for this between-subjects comparison under non-normality; tests stochastic dominance without distributional assumptions.

Lab Modality: Single Lab ($M = 1565$ ms, $Mdn = 1540$ ms) vs. Multi Lab ($M = 1356$ ms, $Mdn = 1331$ ms). $U = 286.0$, $p < .001$, rank-biserial $r = -0.702$ (large). Single Target significantly *slower*. **Reject H_0 .**

Game Modality: Single Game ($M = 2958$ ms, $Mdn = 2888$ ms) vs. Multi Game ($M = 1191$ ms, $Mdn = 1203$ ms). $U = 336.0$, $p < .001$, rank-biserial $r = -1.000$ (complete separation). **Reject H_0 .**

Both modalities show Single Target participants with substantially *higher* RT than Multiple Target - counter-intuitive but likely reflecting task design: Multiple target participants need only identify any one of several targets, reducing effective search set size per response. After outlier removal: Lab $U = 270.0$, $p < .001$, $r = -0.800$; Game $U = 300.0$, $p < .001$, $r = -1.000$.

3.4 Analysis 3 - RQ3: Modality Effect (Wilcoxon Signed-Rank)

Wilcoxon Signed-Rank was applied to the paired (within-subjects) modality comparison under non-normality; tests whether signed-rank differences are symmetric around zero.

Single Target ($n = 21$): Lab $M = 1565$ ms vs. Game $M = 2958$ ms. $W = 0.0$, $p < .001$, $r = 0.000$. All participants were slower in the game; $W = 0$ indicates unanimity of direction. **Reject H_0 .**

Multiple Target ($n = 16$): Lab $M = 1356$ ms vs. Game $M = 1191$ ms. $W = 28.0$,

$p = .039$, $r = 0.206$. Multiple target participants significantly *faster* in the game, suggesting the format is more efficient or engaging for this group. **Reject H_0 .**

After outlier removal: Single ($n = 19$) $W = 0.0$, $p < .001$; Multiple ($n = 14$) $W = 15.0$, $p = .017$. Both remain significant.

3.5 Analysis 4 - RQ4: Effect of Game Level (Friedman Test)

The Friedman test (non-parametric repeated-measures ANOVA analogue for $k \geq 3$ conditions) was used; each participant constitutes a block; game levels are the repeated conditions. Raw trial-level RT is heavily right-skewed, and not all players reached the same maximum level.

Single Target ($n = 21$, Levels 1–15): $\chi^2(14) = 161.93$, $p < .001$. **Reject H_0 .**

Multiple Target ($n = 16$, Levels 1–9): $\chi^2(8) = 38.33$, $p < .001$. **Reject H_0 .**

RT differs significantly across levels in both groups. As shown in Figure 6, Single Target participants show a steep positive RT trajectory with level; Multiple Target participants show a flatter or slightly decreasing trend.

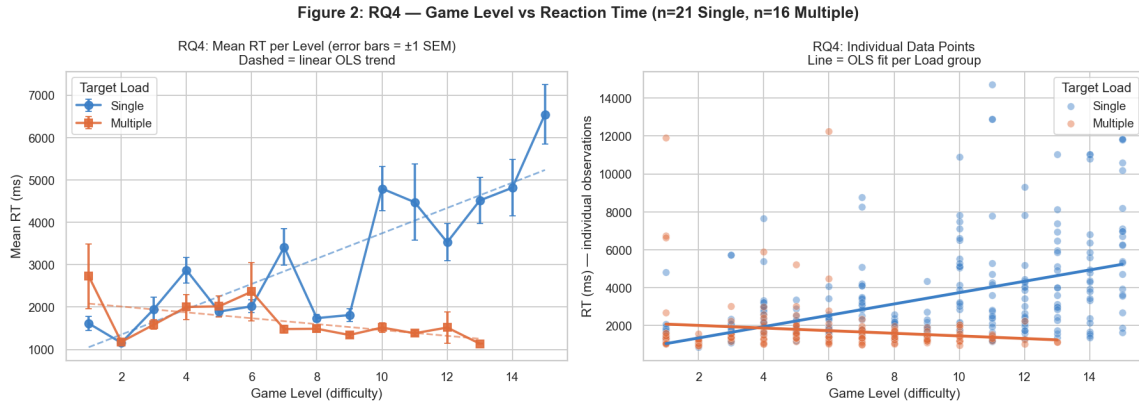


Figure 6: RQ4: Mean RT per game level with OLS trend lines (left) and individual data points with OLS fit (right). Single Target (blue) shows a steep RT increase with level; Multiple Target (orange) remains relatively flat. Error bars = ± 1 SEM.

3.6 Analysis 5 - Mixed ANOVA: Modality $\times$ Target Load

A 2×2 Mixed Factorial ANOVA was applied after confirming $\log(\text{RT})$ normality for all four conditions (Shapiro-Wilk $p > .36$). Load is between-subjects; Modality is within-subjects. Effect sizes reported as partial η^2 .

Main Effect of Target Load: $F(1, 32) = 166.59$, $p < .001$, $\eta_p^2 = 0.839$ (LARGE). **Reject H_0 .**

Main Effect of Modality: $F(1, 32) = 124.52$, $p < .001$, $\eta_p^2 = 0.796$ (LARGE). **Reject H_0 .**

Load × Modality Interaction: $F(1, 32) = 140.82$, $p < .001$, $\eta_p^2 = 0.815$ (LARGE). The modality effect diverges by group: Single Target participants are much slower in game (logRT: 7.33→7.97) while Multiple Target participants are slightly faster (logRT: 7.17→7.08). All three effects exceed the large-effect threshold ($\eta_p^2 > .14$). **Reject H_0 .**

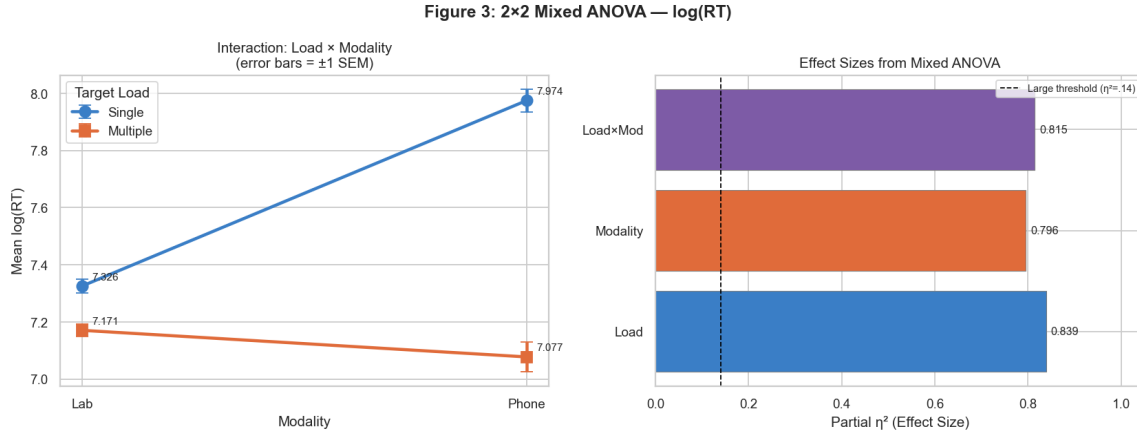


Figure 7: 2×2 Mixed ANOVA on log(RT). Left: interaction plot (Single group increases steeply Lab→Game; Multiple group decreases slightly). Right: partial η^2 for Load (0.839), Modality (0.796), and Load×Modality (0.815).

3.7 Analysis 6 - OLS Regression for Reaction Time (logRT DV)

OLS regression on log(RT) was used to quantify the joint predictive value of Load, Modality, and their interaction. Residual normality confirmed (Shapiro-Wilk $W = 0.982$, $p = .387$). Predictors were mean-centred to reduce multicollinearity.

Model fit: $R^2 = 0.866$, adj. $R^2 = 0.860$, $F(3, 67) = 144.8$, $p < .001$, $N = 71$:

- **Intercept:** $\hat{\beta} = 7.385$, $p < .001$ (grand mean RT ≈ 1627 ms back-transformed).
- **Load (centred):** $\hat{\beta} = 0.534$, $p < .001$; RT ratio $\approx 1.71\times$ (Single vs. Multiple).
- **Modality (centred):** $\hat{\beta} = 0.273$, $p < .001$; RT ratio $\approx 1.31\times$ (Game vs. Lab).
- **Load × Modality:** $\hat{\beta} = 0.758$, $p < .001$; $\approx 2.13\times$ amplification of modality difference for Single vs. Multiple target group.

Residual diagnostics (Figure 8) confirm homoscedasticity and normality; GLM (Gamma, log link) comparison (Figure 9) yields virtually identical multiplicative RT ratios, validating the log-OLS approach.

Figure 4: OLS Regression Diagnostics (log RT model)

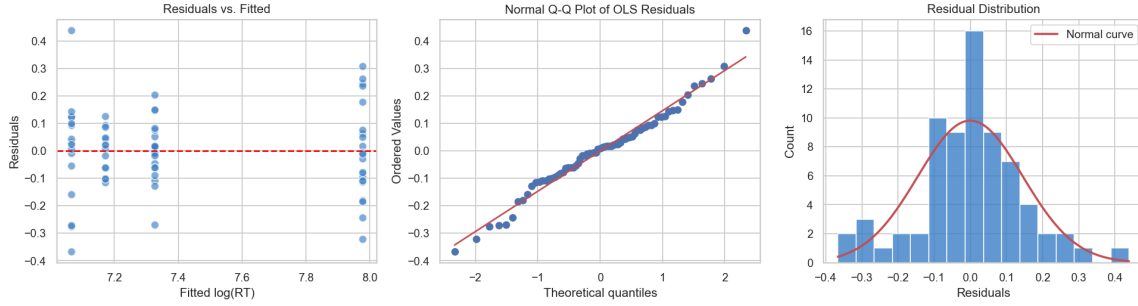


Figure 8: OLS Regression diagnostics for the log(RT) model. Left: Residuals vs. Fitted (homoscedastic). Centre: Normal Q-Q of residuals. Right: Residual distribution with normal curve ($W = 0.982$, $p = .387$).

Figure 5: GLM — Diagnostics and Comparison with OLS

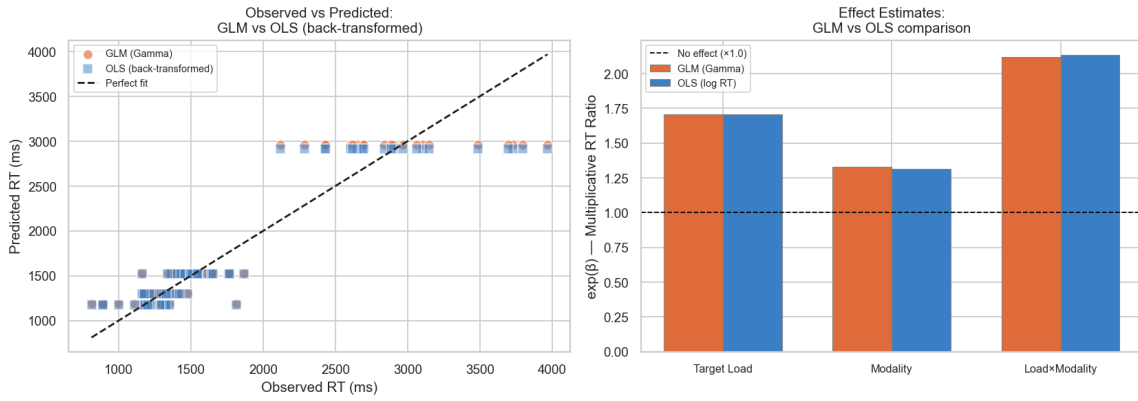


Figure 9: GLM (Gamma, log link) vs. OLS comparison. Left: Observed vs. Predicted RT. Right: $\exp(\hat{\beta})$ for Load, Modality, and Load $\times$ Modality. GLM and OLS give virtually identical effect estimates.

3.8 Analysis 7 - OLS Regression for Complexity Level (Game Level DV)

OLS regression with game level as a continuous predictor and Target Load as a categorical moderator on log(RT) quantifies how RT evolves with difficulty.

Model fit: $R^2 = 0.385$, adj. $R^2 = 0.381$, $F(3, 476) = 99.40$, $p < .001$, $N = 480$:

- **Intercept:** $\hat{\beta} = 7.442$ (Multiple target baseline at Level 1).
- **Load [Single vs. Multiple]:** $\hat{\beta} = -0.308$, $p = .002$. At Level 1, Single target has lower RT than Multiple (controlling for level).
- **Level:** $\hat{\beta} = -0.017$, $p = .156$ (n.s.). No significant level trend for Multiple target participants.
- **Level $\times$ Load[Single]:** $\hat{\beta} = 0.103$, $p < .001$. Each additional level increases Single target log(RT) by 0.103 ($e^{0.103} \approx 1.11$; $\sim 11\%$ per level).

3.9 Analysis 8 - Learning Effect: Multiple Phone First 5 vs. Last 5 (Paired t -test)

A paired t -test was applied; difference-score normality confirmed ($W = 0.930$, $p = .335$). $n = 13$ participants with sufficient data.

First 5 levels: $M = 2029.3$ ms, $SD = 769.1$ ms. Last 5 levels: $M = 1902.3$ ms, $SD = 506.1$ ms. Difference: -127.0 ms. $t(12) = 0.560$, $p = .586$. **Fail to Reject H_0 .**

A numerical RT decrease (~ 127 ms) is present but not statistically significant. Large between-participant variability and the small sample ($n = 13$) likely limit power to detect this effect.

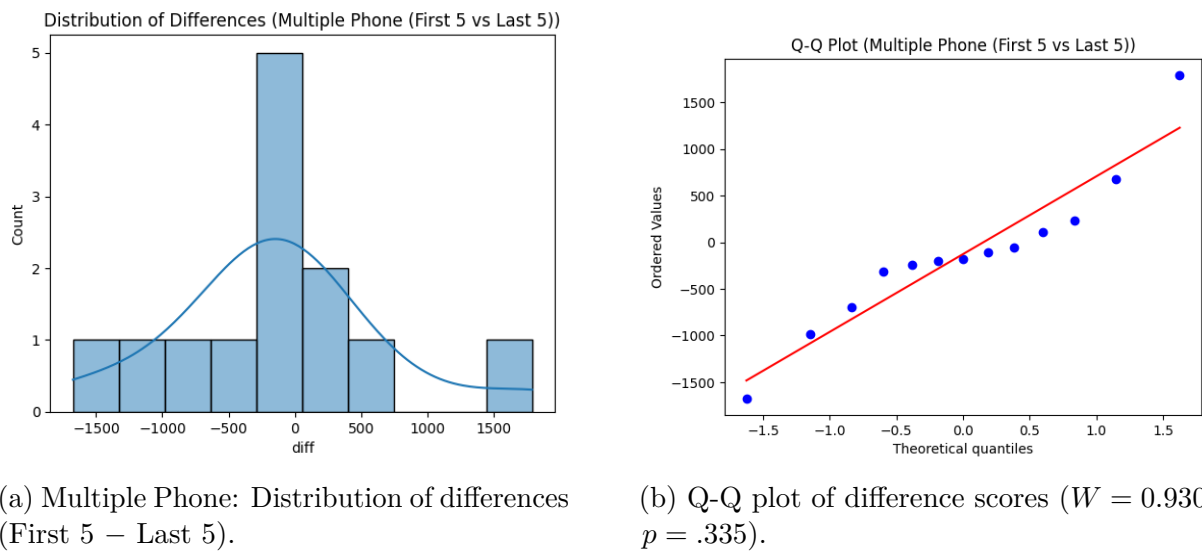


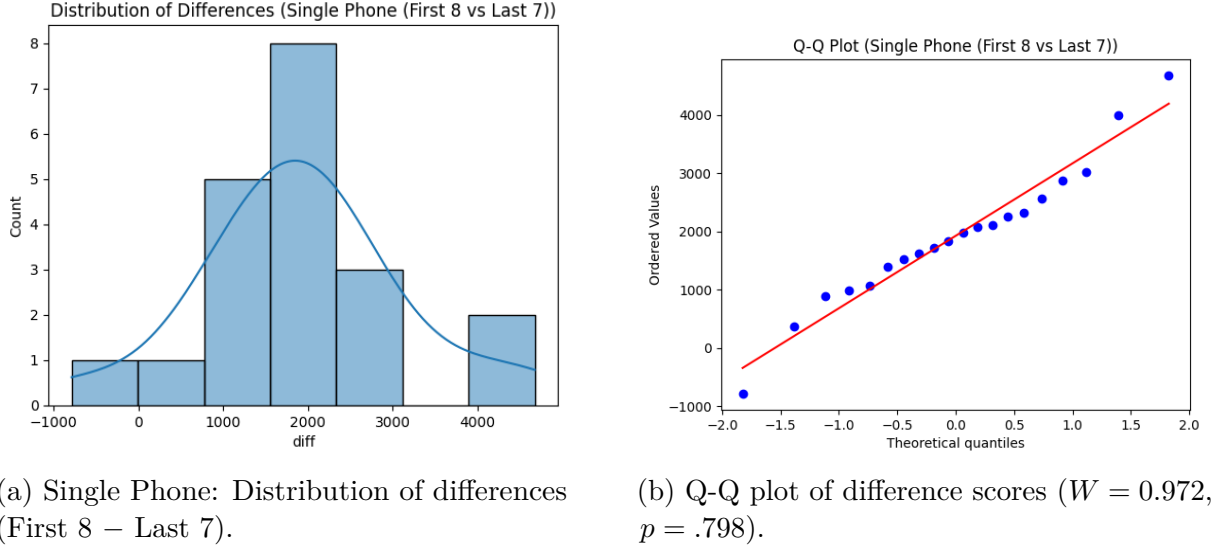
Figure 10: Multiple Phone paired t -test diagnostics (Analysis 8).

3.10 Analysis 9 - Complexity Effect: Single Phone First 8 vs. Last 7 (Paired t -test)

Paired t -test applied after confirming difference-score normality ($W = 0.972$, $p = .798$). $n = 20$ participants.

First 8 levels: $M = 2031.2$ ms, $SD = 462.0$ ms. Last 7 levels: $M = 3954.3$ ms, $SD = 1056.3$ ms. Difference: $+1923.1$ ms. $t(19) = -7.123$, $p < .001$. **Reject H_0 .**

Single Phone participants were significantly *slower* in later levels, indicating a strong **difficulty/fatigue effect** (~ 1923 ms RT increase). Combined with the OLS level effect (Analysis 7 - $\sim 11\%$ per level), this confirms that increasing game complexity drives substantially higher RT for Single Target participants.

Figure 11: Single Phone paired t -test diagnostics (Analysis 9).

4 Conclusions and Future Directions

4.1 Summary of Findings

This study evaluated a smartphone-based Selective Attention Game as a cognitive assessment instrument by comparing it to a standard laboratory paradigm. The principal findings, taken together, present a nuanced picture of the game’s validity and utility.

Concurrent validity (RQ1) was not supported within either target-load group individually (Single: $\rho = 0.396$, $p = .076$; Multiple: $\rho = 0.035$, $p = .897$). This suggests that game and lab RTs do not reliably rank participants in the same order within groups, limiting the instrument’s criterion validity at this sample size.

Target load effects (RQ2) were highly significant ($p < .001$) and counter-intuitive: Single Target participants were consistently *slower* than Multiple Target participants. This pattern is most likely an artefact of game design, where the single-target condition requires more precise and sustained search than the multiple-target condition.

Modality effects (RQ3) confirmed that the game and lab tasks produce meaningfully different RT profiles ($p < .001$ for Single group; $p = .039$ for Multiple group), with the game eliciting dramatically slower RT in Single target participants but slightly faster RT in Multiple target participants. The large Load $\times$ Modality interaction from the Mixed ANOVA ($\eta_p^2 = 0.815$) replicates this finding, indicating that the game does not provide a simple linear rescaling of lab performance.

Game level effects (RQ4) were significant ($p < .001$ in both groups), and the OLS regression confirmed that each level increment increases Single target RT by $\sim 11\%$ — a strong difficulty trajectory. Paired t -tests further revealed that this manifests as a fatigue/difficulty effect for Single phone users (RT increased by ~ 1923 ms from early to late levels, $p < .001$), while Multiple phone users showed no significant learning or fatigue effect.

Table 2: Comprehensive Summary of All Nine Statistical Analyses

Analysis	Test	Key Statistic	One-Line Conclusion
RQ1 Concurrent Validity	Spearman ρ	Single: $\rho = 0.396$, $p = .076$ Multi: $\rho = 0.035$, $p = .897$ Combined: $\rho = 0.621$, $p < .001$	Within-group validity not supported; combined correlation likely inflated by group-level RT differences.
RQ2 Target Load Effect	Mann-Whitney U	Lab: $U = 286$, $p < .001$, $r = -0.70$ Game: $U = 336$, $p < .001$, $r = -1.00$	Single target participants significantly <i>slower</i> than Multiple in both modalities; counter-intuitive load effect.
RQ3 Modality Effect	Wilcoxon Signed-Rank	Single: $W = 0$, $p < .001$ Multi: $W = 28$, $p = .039$	Modality significantly alters RT in both groups: Single slower in game; Multiple faster in game.
RQ4 Game Level Effect	Friedman Test	Single: $\chi^2(14) = 161.93$, $p < .001$ Multi: $\chi^2(8) = 38.33$, $p < .001$	RT differs significantly across game levels in both groups, confirming a systematic level effect.
ANOVA Mixed Design	2×2 Mixed ANOVA on log(RT)	Load: $F(1, 32) = 166.6$, $\eta_p^2 = 0.839$ Modality: $F(1, 32) = 124.5$, $\eta_p^2 = 0.796$ Interaction: $F(1, 32) = 140.8$, $\eta_p^2 = 0.815$	All three effects significant with very large effect sizes; interaction reveals diverging modality effects by group.
OLS Regression (RT)	OLS on log(RT)	$R^2 = 0.866$, $F(3, 67) = 144.8$, $p < .001$; interaction $\hat{\beta} = 0.758$, $p < .001$	Load, Modality, and interaction explain 86.6% of RT variance; game amplifies RT differences between load groups.
OLS Regression (Level)	OLS on log(RT) $\sim$ Level $\times$ Load	$R^2 = 0.385$, $F(3, 476) = 99.40$, $p < .001$; Level $\times$ Single: $\hat{\beta} = 0.103$, $p < .001$	Each additional game level increases Single target RT by $\sim 11\%$; no significant level effect for Multiple.
Learning Effect MP (5 vs. 5)	Paired t -test	$t(12) = 0.560$, $p = .586$	No significant learning effect for Multiple Phone; RT decreased by ~ 127 ms but not significantly.
Complexity SP (8 vs. 7)	Paired t -test	$t(19) = -7.123$, $p < .001$	Strong difficulty/fatigue effect: Single Phone RT increased by ~ 1923 ms in later levels ($p < .001$).

4.2 Limitations

- Sample sizes ($n = 16$, $n = 21$) limit statistical power, particularly for detecting within-group validity correlations.
- The near-ceiling accuracy for most conditions (especially Single Lab: $M = 100\%$) means accuracy cannot discriminate between participants, reducing the available information about performance quality.
- Counterbalancing of modality order was applied, but carry-over effects were not formally tested.

4.3 Future Directions

1. **Revised game calibration:** Adjusting game difficulty scaling so that Single and Multiple target conditions produce comparable RT ranges would facilitate direct cross-group validity comparisons.
2. **Reliability testing:** Test-retest reliability over multiple sessions would establish the game's temporal stability, a prerequisite for clinical use.
3. **Broader cognitive battery:** Correlating game scores with other validated attention measures (e.g., ADHD-Q, BRIEF) would strengthen construct validity beyond RT-to-RT comparison.

4.4 Code and Contributions

Team Contributions		
Member	Roll No.	Contribution
Pranav	2025201021	EDA, RQ4, OLS ($RT_{\log}$), OLS (Level)
Vaibhav	2025201067	RQ2, RQ3, Mixed ANOVA
Afzal Basha	2025201097	Design, RQ1, Paired t -tests, Report

GitHub Repository: [GitHub Link](#)

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